Software Requirements Specification and Design for KisanSetu

Version 1.0

Project: KisanSetu: Farmer-Centric Digital Agri-Ecosystem Platform

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Revision history

Name	Date	Reason for Changes	Version

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1. Introduction

1.1. Purpose

The purpose of this document is to provide a detailed and structured technical overview of KisanSetu, a digital agri-ecosystem platform designed to address long-standing challenges in the Indian agricultural sector. It clearly explains the core functionalities of the system, the user interfaces through which farmers, buyers, and administrators interact with the platform, and the operational limitations within which the system is designed to function. The document also outlines how these features work together to improve market access, transparency, and efficiency for farmers. This Software Requirements Specification (SRS) serves as a reference for all stakeholders, including small and marginal farmers, government bodies, and policymakers, as well as the development team responsible for designing, building, and maintaining the system.

1.2. Intended Audience and Reading Suggestions.

Intended Audience

- **Farmers :** The central focus of the platform. This includes smallholder farmers seeking fair pricing, affordable equipment rentals, and professional agronomy advice to increase productivity.
- **Retail and Wholesale Buyers:** Individuals or businesses looking for a transparent marketplace to source high-quality, fresh produce directly from the farm with end-to-end traceability.
- **AgriTech Entrepreneurs and Startups:** Individuals looking for a strategic roadmap to address market inefficiencies, minimize post-harvest waste, and build scalable digital-physical ("phygital") business models.
- **Government Officials and Policy Makers:** Stakeholders focused on the integration of digital platforms with existing schemes like e-NAM, the Soil Health Card Scheme, and Farmer Producer Organizations (FPOs).
- **Investors and Venture Capitalists:** Those seeking an overview of the Indian AgriTech market potential, competitive landscape, and the economic viability of multi-sided digital platforms.
- **Agricultural Professionals and FPOs:** Experts and collective farmer groups interested in how technology can democratize access to machinery, professional expertise, and transparent market pricing.

Reading Suggestions by Interest

If you are interested in...	Focus on these Sections
Direct Market Access & Pricing	Section 1.3.1: Explains the farmer-driven pricing model and the direct-to-buyer marketplace.
Operational Efficiency & Equipment	Section 1.3.2: Details the on-demand hiring of machinery and professional experts.`
Platform Trust & Accessibility	Section 4: Discusses bridging the digital divide through on-ground "phygital" hubs and simple UI/UX.
Growth & Future Roadmap	Section 4: Provides the timeline for moving from a regional pilot to a national agri-ecosystem.

1.3. Scope of Project

KisanSetu is an integrated software platform developed to enhance farmer productivity and significantly improve their income by digitally streamlining key agricultural activities. It provides automated tools that simplify access to agricultural markets, enable affordable rental of farming equipment, and support efficient post-harvest management processes. By directly connecting farmers with buyers and service providers, the system reduces reliance on traditional intermediaries who typically absorb a large share of the final consumer price. This direct and transparent approach helps farmers receive fair value for their produce, with the overall objective of increasing farmer income by approximately 25–35 percent while promoting efficiency and sustainability across the agricultural value chain.

Specifically, the project integrates:

1.3.1. Smart Marketplaces (Real-Time Mandi Pricing & Direct B2B/B2C Sales)

The Smart Marketplace module enables farmers to directly access agricultural markets using digital platforms. It provides **real-time mandi prices** from multiple locations, allowing farmers to make informed decisions about when and where to sell their produce. Instead of relying on local middlemen, farmers can compare prices across mandis and choose the most profitable option.

Additionally, this module supports **direct selling models**:

- **B2B (Business-to-Business):** Farmers can sell directly to wholesalers, food processors, retailers, or exporters.

- **B2C (Business-to-Consumer):** Farmers can sell produce directly to consumers through online listings or local delivery networks.

By removing intermediaries, farmers receive a larger share of the final consumer price, improving price transparency, reducing exploitation, and increasing overall income.

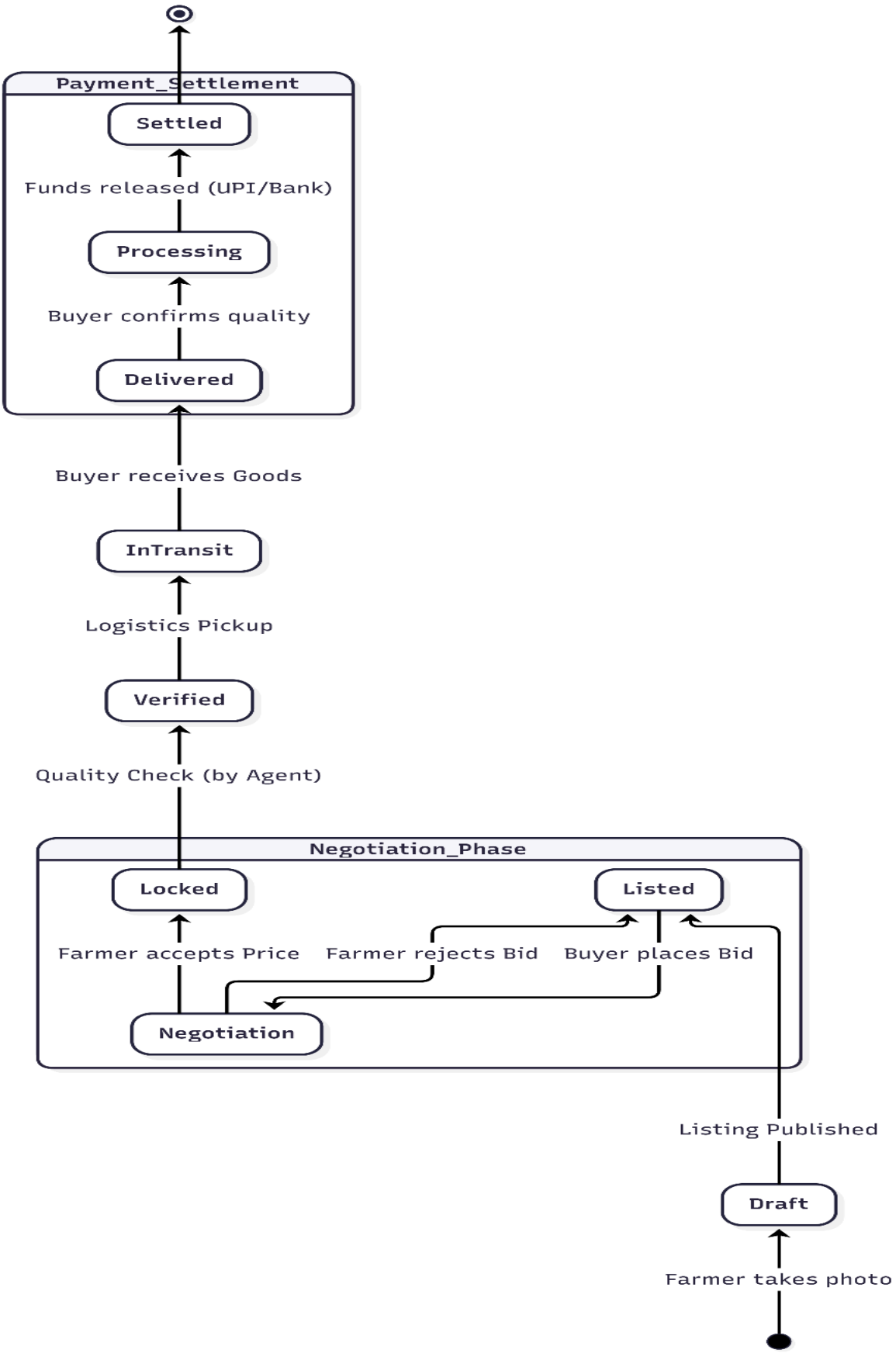

Fig 1 : State Transition Diagram for the Crop Listing and Negotiation Module.

1.3.2. Asset-Sharing Models (Affordable Rental of Modern Machinery)

The Asset-Sharing Model focuses on solving the problem of **high capital costs of modern agricultural equipment**. Many small and marginal farmers cannot afford machinery such as tractors, harvesters, seed drills, or agricultural drones.

KisanSetu provides a **rental-based digital platform** where:

- Equipment owners (government agencies, cooperatives, or private owners) can list machinery.
- Farmers can book machinery on an hourly, daily, or seasonal basis at affordable rates.

This model ensures:

- Access to advanced technology without ownership costs
- Reduced dependency on manual labor
- Timely farming operations, leading to higher productivity

By democratizing access to modern tools, the system helps farmers improve efficiency, crop quality, and yield.

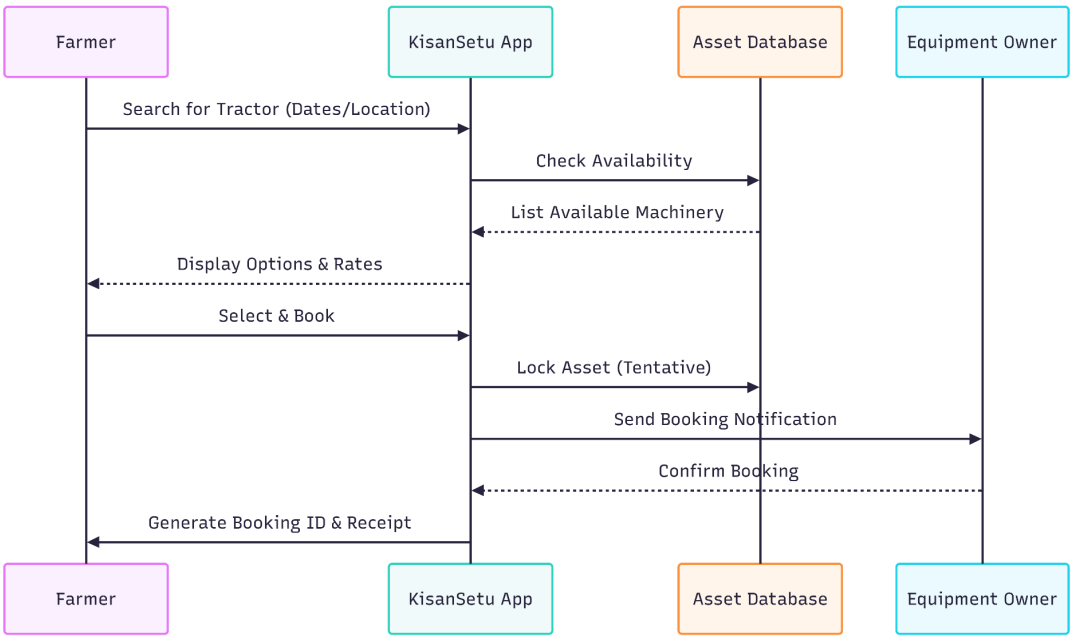

Fig 2 : Step-by-step communication between the Farmer, the App, and the Equipment Owner's Database.

1.3.3. Government Service Integration (eNAM, Soil Health Cards, etc.)

KisanSetu integrates seamlessly with existing **government agricultural platforms** to provide farmers with centralized access to public services. This reduces paperwork, delays, and lack of awareness.

Key integrations include:

- **eNAM (Electronic National Agriculture Market):** Enables farmers to participate in a nationwide digital market, ensuring better price discovery and wider market access.

- **Soil Health Card System:** Provides data-driven recommendations on soil nutrients, helping farmers choose the right crops and fertilizers, reducing input costs and environmental damage.

- Other schemes related to subsidies, crop insurance, and advisories can also be linked.

This integration ensures that farmers receive **personalized, data-backed guidance** while fully utilizing government support schemes, leading to better decision-making and sustainable farming practices.

1.4. References

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2. Overall Description

2.1. Project Perspective

- Kisan Setu is an integrated e-commerce platform designed to establish a direct digital nexus between agricultural producers and local purchasers, thereby cultivating a transparent and highly efficient agricultural marketplace.

- The platform fundamentally reconfigures the conventional, multi-layered distribution infrastructure by employing a Peer-to-Peer (P2P) and Business-to-Business (B2B) operational paradigm. This allows farmers to engage directly with a diverse range of buyers, including retailers, wholesalers, and institutional entities.

- By granting direct market access, Kisan Setu facilitates a fair and accurate price discovery mechanism, which is intrinsically linked to local supply-and-demand dynamics. This yields the dual benefit of securing enhanced financial returns for farmers while concurrently reducing procurement expenditures for buyers.

- Furthermore, the platform actively promotes the digitization of agricultural processes. This advancement leads to substantial improvements in planning efficacy and a significant reduction in post-harvest losses, largely attributable to superior visibility regarding produce availability.

- In addition, Kisan Setu integrates a dedicated feature for the rental and sale of agricultural machinery, enabling both farmers and sellers to list specialized equipment. This functionality optimizes resource utilization and contributes directly to an overall augmentation of farm productivity.

2.2. Key Functionalities

Smart Digital Marketplace and Price Discovery
The platform provides farmers with access to real-time mandi prices from multiple locations, enabling transparent price discovery. Farmers can compare market rates and choose the most profitable selling option, ensuring fair valuation of their produce.

Direct B2B and B2C Trading Mechanisms
KisanSetu enables direct transactions between farmers and buyers through B2B

and B2C models. This removes intermediaries such as commission agents, allowing farmers to receive a higher share of the final consumer price while reducing costs for buyers.

Agricultural Equipment Rental and Asset Sharing
The platform offers an asset-sharing model where agricultural machinery owners can list equipment for rent or sale. Farmers can access modern tools on flexible rental terms, reducing capital investment and improving operational efficiency.

Post-Harvest Management Support
KisanSetu improves post-harvest management by facilitating better coordination between farmers and buyers. This reduces crop wastage and improves storage planning.

Integration with Government Agricultural Services
The system integrates with key government platforms such as eNAM and the Soil Health Card system. This allows farmers to participate in national markets, receive soil-based crop recommendations, and access government-backed agricultural data and services.

Income Enhancement and Sustainable Agriculture
By combining digital market access, equipment sharing, and government service integration, the platform aims to increase farmer income by approximately 25–35 percent. It also promotes sustainable farming practices through efficient resource utilization and data-driven decision-making.

2.3. User Classes and Characteristics

To ensure the success of our "phygital" (physical + digital) platform, it is critical to understand the specific needs and behaviors of the people who will use it. Our solution is designed to bridge the gap between technology and traditional farming by catering to four distinct user groups. Below is a detailed breakdown of these users and how our system addresses their unique requirements.

2.3.1. Smallholder Farmers (The Core Producers)

- **Who They Are:** This group makes up the majority of our user base, representing about 80% of India's landholders. They are the backbone of the agricultural sector but often operate with limited resources.
- **Key Characteristics:**
    - **Limited Digital Skills:** Many are not comfortable with complex smartphone apps or typing on screens.

    - **High Skepticism:** They are often wary of new technology and need proof that a system works before trusting it.
    - **Language Diversity:** They speak various local dialects and may not be fluent in English or even standard Hindi.
- **Our Solution for Them:**
    - **Voice-First Interface:** Instead of typing, farmers can use voice commands to log data or ask questions.
    - **Simple Visuals:** The app will use icons and pictures rather than heavy text to make navigation intuitive.
    - **Trust Building:** The system will provide transparent pricing and immediate confirmations to build confidence in the platform.

2.3.2. Commercial Buyers (Retailers & Wholesalers)

- **Who They Are:** These are the businesses and individuals who purchase produce in bulk to sell to end consumers. They include supermarket chains, local grocery stores (kiranas), and food processing companies.
- **Key Characteristics:**
    - **Quality Conscious:** Their reputation depends on selling fresh, high-quality food.
    - **Speed & Efficiency:** They need to make purchasing decisions quickly and require reliable delivery timelines.
    - **Demand for Transparency:** They want to know exactly where the produce comes from to ensure safety and quality.
- **Our Solution for Them:**
    - **Traceability:** A "digital passport" for every batch of produce, showing its journey from farm to fork. This guarantees origin and quality.
    - **Real-Time Dashboard:** A clear view of available stock, expected harvest dates, and transparent pricing to help them plan their inventory.

2.3.3. Micro-Entrepreneurs (Field Agents / "The Human Touch")

- **Who They Are:** Modeled after successful networks like DeHaat, these are local individuals who act as the bridge between the digital platform and the physical world of the village.
- **Key Characteristics:**
    - **Tech-Savvy:** They are comfortable using smartphones and digital tools.
    - **Community Trusted:** They are locals who are known and respected by the farmers.
    - **Multi-Taskers:** Their role involves onboarding farmers, training them on the app, and managing the initial collection of produce.

- **Our Solution for Them:**
    - **Agent App:** A dedicated tool that allows them to register new farmers and schedule training at local hubs.
    - **Training Modules:** Digital resources to help them educate farmers on best practices and platform usage.

2.3.4. Agricultural Specialists (The Experts)

- **Who They Are:** Agronomists, soil scientists, and other agricultural professionals who provide expert advice and consulting services.
- **Key Characteristics:**
    - **Knowledge-Driven:** They have specialized expertise that can significantly improve crop yields and health.
    - **Project-Based Work:** They often work on specific issues (e.g., pest control, soil health) for a defined period.
- **Our Solution for Them:**
    - **Consultant Dashboard:** A professional interface where they can list their services, manage appointments with farmers or cooperatives, and track their advisory history.
    - **Remote Diagnostics:** Tools that allow them to view crop photos or data uploaded by farmers to provide accurate advice without always needing a site visit.

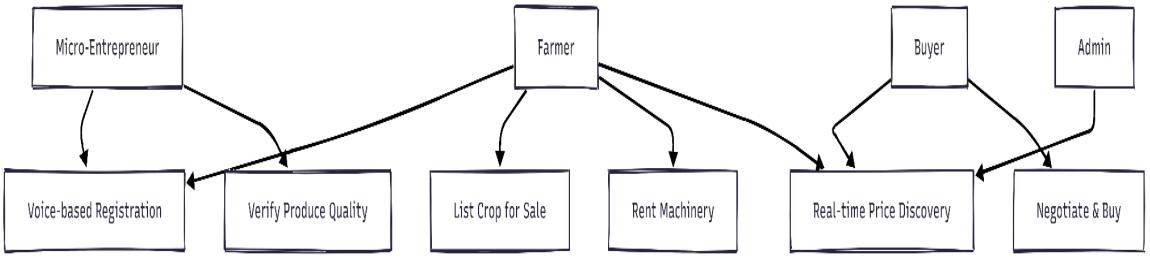


Fig 3: Defines the scope of what each user can do within the system.

2.4. Operating Environment

The platform must be resilient enough to operate in the challenging conditions of rural India:

- **Client-Side:** The primary interface is a **lightweight platform**. Given that 74.8% of rural households own smartphones, the platform must be optimized for mid-to-low-range hardware with limited storage.
- **Connectivity:** The environment is characterized by intermittent internet access. The app must support **offline data entry** and synchronization once a stable connection is restored.

- **Backend Infrastructure:** A cloud-based architecture is required to process massive amounts of data for AI-driven demand prediction .

2.5. Dependencies

The successful operation and feature realization of the KisanSetu platform are contingent upon several external systems and operational factors:

1. **Stable Internet/Mobile Connectivity in Rural Areas:** While the application supports offline data entry, continuous, stable connectivity is necessary for real-time price discovery, transaction execution, and backend synchronization. *The platform's real-time features are fully dependent on the availability of this infrastructure.*

2. **API Access to Government Platforms:** Full integration and functionality of ***Section 1.3.3*** and the related System Feature depend on continuous, reliable API access to key government platforms such as eNAM and the Soil Health Card system for fetching data and enabling farmer participation.

3. **Third-Party Payment Gateway Integration:** The secure and efficient execution of B2B and B2C transactions requires integration with a reliable and regulated payment gateway (e.g., UPI, Net Banking services) that can handle large transaction volumes and is compliant with Indian financial regulations.

4. **Adoption and Training by Micro-Entrepreneurs/Field Agents:** The success of the "phygital" model ***(Section 2.3)*** heavily depends on the active recruitment, successful training, and sustained participation of local micro-entrepreneurs who serve as the crucial human interface for farmer onboarding and first-mile logistics.

5. **Data Quality and Validation from Mandis:** The core function of the Real-Time Mandi Price Dashboard ***(Section 4)*** is dependent on receiving timely, accurate, and standardized pricing data from various agricultural markets (mandis).

3. External Interface Requirements

3.1. User Interfaces (UI)

The user interface must be designed with extreme simplicity to address the fact that 23.4% of rural adults lack basic digital literacy. The UI will be categorized by the specific needs of each user class.

3.1.1. Farmer Interface

- **Multilingual Support:** The interface must support major Indian regional languages and use clear, simple visuals rather than dense text to accommodate varying literacy levels.
- **Voice-Enabled Search:** Integration of voice commands to allow farmers to search for crop prices or equipment rentals hands-free.
- **Icon-Driven Navigation:** Core functions like "Sell Produce," "Rent Tractor," or "Soil Test" will be represented by large, high-contrast icons to improve usability.
- **Offline Indicator:** A clear visual cue (e.g., a green/red sync icon) to show if the platform is currently synced with the central database or operating in offline mode.

3.1.2. Buyer Interface

- **Real-Time Price Dashboard:** A data-rich interface showing live pricing from over 7,000 agricultural mandis across India.
- **Negotiation Status:**A phone-verified deal indicator where farmers update their intent to sell after direct contact with a buyer. The updated status is reflected on the buyer’s dashboard, ensuring transparent and farmer-driven deal progression.

3.1.3. Micro-Entrepreneur/Admin Dashboard

- **Onboarding Wizard:** A simplified registration tool for field agents to quickly upload farmer data and government ID information.

3.2. Hardware Interface

Since the platform relies on a "phygital" model, the software must interface seamlessly with various hardware components to ensure operational efficiency.

- **Soil Testing Equipment:** Interfaces to accept digital data outputs from government soil testing labs, such as those at Krishi Vigyan Kendra.

- **Biometric/Mobile Sensors:** Utilization of smartphone cameras for scanning QR codes on specialized packaging (PCM technology) and for biometric verification during financial settlements.
- **Handheld POS Devices:** Compatibility with portable printers or tablets used by micro-entrepreneurs at physical hubs for generating transaction receipts.

3.3. Software Interface

The platform acts as a "value-added layer" that must communicate with external software ecosystems to provide a holistic solution.

- **Government Platform APIs:**
    - **e-NAM API:** Integration to pull real-time auction data and wholesale price discovery information.
    - **Soil Health Card Database:** A secure interface to retrieve soil parameters and fertilizer recommendations.
    - **FARMS App Integration:** Potential data partnerships to sync equipment availability with the platform’s rental marketplace.
- **Blockchain Protocol:** Software hooks to a decentralized ledger to record tamper-proof transaction and movement data, ensuring regulatory compliance.
- **Payment Gateways:** Integration with UPI and formal banking APIs to ensure timely, secure, and transparent online payment settlements for farmers.

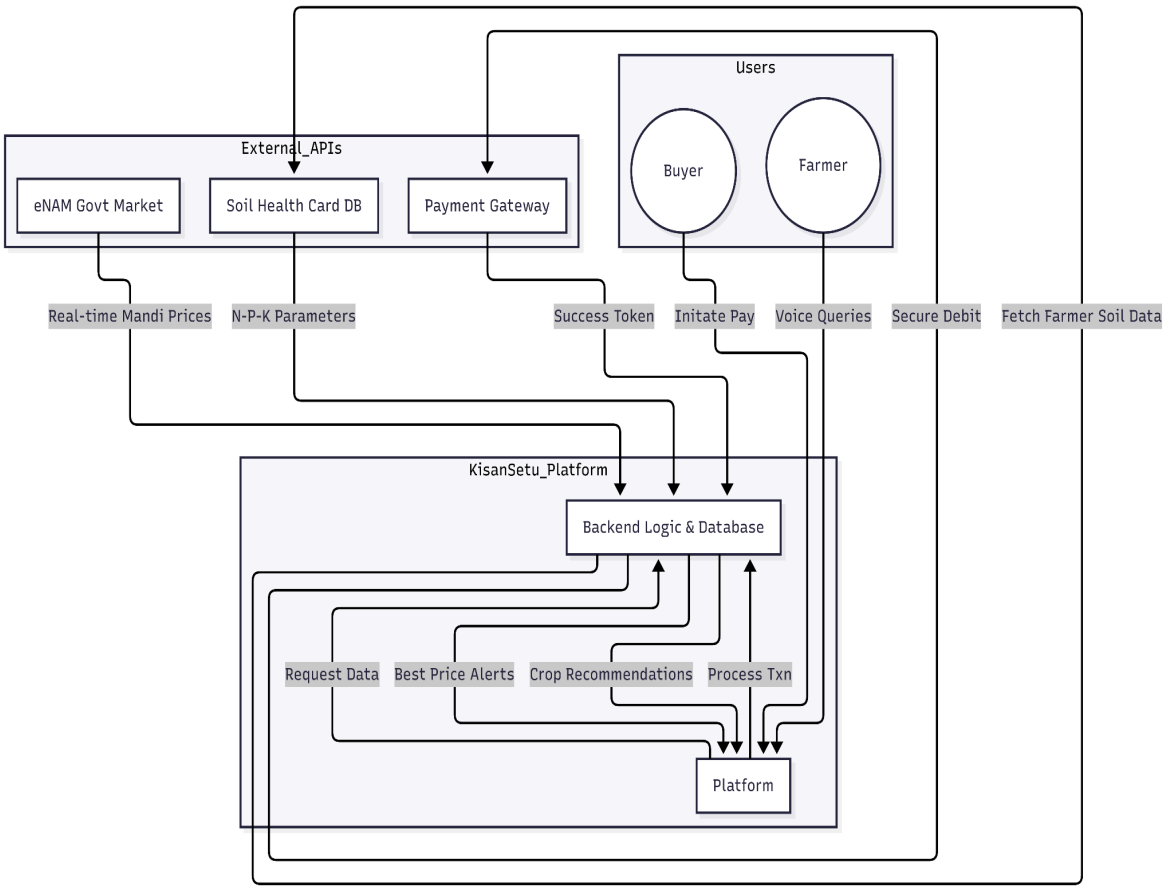


Fig 4 : Data Flow Diagram (DFD) showing integration with external Government and Financial APIs.

3.4. Communication Interface

Given the rural operating environment, the communication protocols must be robust and adaptable to varying network conditions.

- **Network Protocols:** Support for HTTP/HTTPS for secure data transfer, and MQTT (Message Queuing Telemetry Transport) for lightweight, low-bandwidth IoT sensor communication.
- **Synchronization Logic:** A store-and-forward mechanism that allows the platform to function in 2G/3G zones and automatically sync data once 4G/5G or Wi-Fi becomes available.
- **SMS/USSD Gateway:** A fallback communication channel to send price alerts and transaction confirmations via SMS to users without consistent data access.
- **Push Notifications:** Low-latency notification system for urgent alerts, such as temperature threshold breaches in the cold chain or new buyer bids.
- **Cloud Hosting:** Secure communication between the mobile client and centralized cloud servers for AI-driven demand prediction and logistics forecasting.

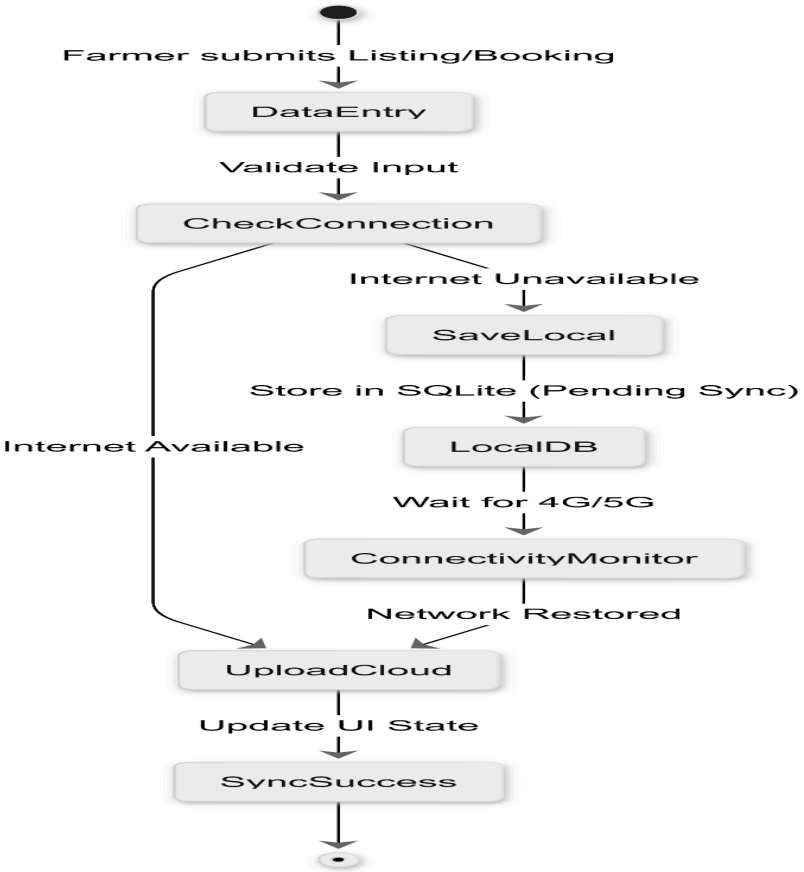


Fig 5 : System's technical resilience.

4. System Features

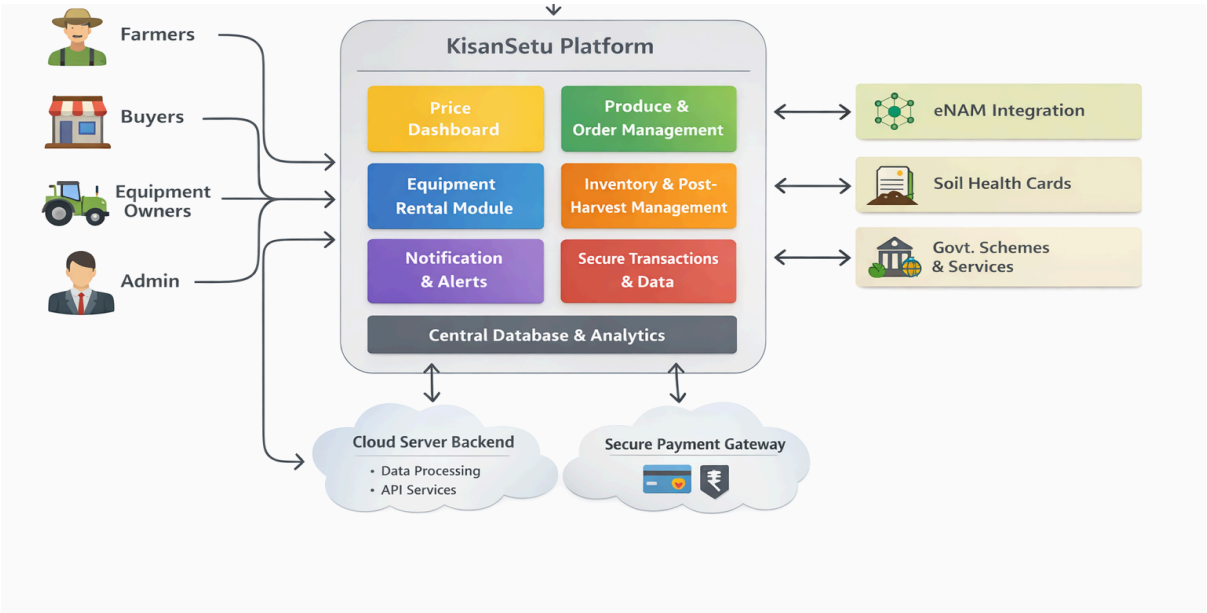


Fig 6 : Role -based user access and management

This section outlines the specific functional requirements of the **KisanSetu** platform. Each feature is detailed by its input, processing logic, and expected output to ensure unambiguous development guidelines.

4.1 Role-Based User Access and Management (RBAC)

4.1.1 Description

The system shall implement a robust Role-Based Access Control (RBAC) mechanism. This ensures that the platform is secure and that users (Farmers, Buyers, Equipment Owners, and Administrators) only interact with interfaces and data relevant to their specific workflow. This module handles authentication, authorization, and profile management.

4.1.2 Functional Requirements

- **FR-1.1:** The system shall allow users to register using a mobile number (with OTP verification) or email address.
- **FR-1.2:** During registration, the user must select a primary role: Farmer, Buyer, or Equipment Owner.
- **FR-1.3:** The **Farmer** role shall have access to produce listing, inventory management, and equipment booking interfaces.
- **FR-1.4:** The **Buyer** role shall have access to the marketplace, bidding systems, and order history.

- **FR-1.5:** The **Equipment Owner** role shall have access to asset management, rental calendars, and earning reports.
- **FR-1.6:** The **Administrator** shall have a "Super User" view to moderate listings, resolve disputes, and manage user accounts.
- **FR-1.7:** The system must encrypt user passwords using hashing algorithms (e.g., decrypt) before storage.

4.1.3 Processing Logic

- **Input:** User credentials, Role selection, KYC documents (Aadhaar/PAN if applicable).
- **Processing:** Validate format -> Verify OTP -> Check database for duplicates -> Create User Object -> Assign Permission Token.
- **Output:** Successful login redirects to the role-specific dashboard.

4.2 Real-Time Mandi Price Dashboard

4.2.1 Description

A centralized data visualization module that aggregates and displays live market prices from various APMCs (Agricultural Produce Market Committees) and local mandis. This feature empowers farmers to decide *when* and *where* to sell.

4.2.2 Functional Requirements

- **FR-2.1:** The system shall fetch data from government APIs (or simulated internal databases) regarding daily crop prices.
- **FR-2.2:** Users shall be able to filter price data by **State**, **District**, **Commodity** (Crop), and **Date**.
- **FR-2.3:** The dashboard must display price trends using graphical representations (Line charts for price history over the last 30 days).
- **FR-2.4:** The system shall highlight the "Best Price Available" for a selected crop within a 50km radius of the user’s location.
- **FR-2.5:** The data must refresh at a defined interval (e.g., every 4 hours) to ensure accuracy.

4.2.3 Processing Logic

- **Input:** User location data, Selected Crop filter.
- **Processing:** Query external API -> Parse JSON response -> Calculate average/min/max prices -> Render visualization components.
- **Output:** Interactive graphs and tables displaying current market rates.

4.3 Direct Produce Listing and Transaction System

4.3.1 Description

This is the core e-commerce engine of KisanSetu. It facilitates the B2B (Business to Business) and B2C (Business to Consumer) exchange of goods. It allows farmers to create digital catalogs of their harvest and allows buyers to procure them via fixed price or bidding.

4.3.2 Functional Requirements

- **FR-3.1:** Farmers shall be able to create a listing including: Crop Name, Variety, Quantity (in kg/quintal), Expected Price, Harvest Date, and Images.
- **FR-3.2:** The system shall support two transaction modes: **"Buy Now"** (Fixed Price) and **"Make an Offer"** (Bidding/Negotiation).
- **FR-3.3:** Buyers shall be able to add multiple items to a cart and proceed to checkout.
- **FR-3.4:** The system shall integrate a secure Payment Gateway for holding funds in escrow until the buyer confirms receipt of goods.
- **FR-3.5:** Upon successful transaction, the listing quantity must automatically decrement in the database.

4.3.3 Processing Logic

- **Input:** Product details (Farmer), Order placement (Buyer).
- **Processing:** Validate stock availability -> Lock stock during checkout -> Process Payment -> Generate Invoice -> Notify Seller.
- **Output:** Order Confirmation ID, Digital Invoice, and Updated Inventory.

4.4 Agricultural Equipment Rental and Booking Module

4.4.1 Description

A "Sharing Economy" feature allowing equipment owners to monetize idle machinery (tractors, harvesters, tillers) and providing small-scale farmers access to affordable mechanization without capital investment.

4.4.2 Functional Requirements

- **FR-4.1:** Equipment Owners shall be able to list machinery with specifications (HP, Model), rental cost (per hour/day), and location.

- **FR-4.2:** The system shall provide a **Calendar Interface** showing the availability status of each asset.
- **FR-4.3:** Farmers shall be able to request a booking for specific dates/times.
- **FR-4.4:** The Owner must have the ability to Approve or Reject a booking request within a stipulated time (e.g., 24 hours).
- **FR-4.5:** The system shall calculate total rental cost based on duration and base rates.

4.4.3 Processing Logic

- **Input:** Asset ID, Start Date/Time, End Date/Time.
- **Processing:** Check overlap in Booking Table -> Calculate Cost -> Send Request to Owner -> Update Status to 'Pending'.
- **Output:** Booking Status Notification and Rental Agreement generation.

4.5 Inventory and Post-Harvest Management Support

4.5.1 Description

A digital ledger system for farmers to track what they have harvested, what is currently in storage, and the shelf-life status of their produce. This assists in reducing post-harvest losses.

4.5.2 Functional Requirements

- **FR-5.1:** The system shall allow manual entry of harvest batches, including crop type, quantity, and storage location (e.g., Warehouse A, Cold Storage).
- **FR-5.2:** The system shall estimate the expiry date based on standard shelf-life data for the specific crop variety.
- **FR-5.3:** Users shall receive alerts when a batch is nearing its expiration date.
- **FR-5.4:** The system shall generate a "Stock Summary Report" downloadable in PDF format.

4.5.3 Processing Logic

- **Input:** Batch entry details.
- **Processing:** Timestamp creation -> Shelf-life calculation -> Database insertion.
- **Output:** Visual inventory levels (Low/Medium/High) and Expiry Alerts.

4.6 Government Service Integration Interface

4.6.1 Description

To bridge the gap between farmers and national schemes, this module acts as an API gateway, fetching and displaying relevant data from platforms like eNAM (National Agriculture Market) and Soil Health Card databases.

4.6.2 Functional Requirements

- **FR-6.1:** The system shall provide an interface for farmers to input their Soil Health Card ID.
- **FR-6.2:** Based on Soil Health data (N-P-K values), the system shall recommend suitable crops or fertilizers (using a rule-based recommendation engine).
- **FR-6.3:** The system shall display a news feed of current government subsidies and schemes relevant to the user's state.
- **FR-6.4:** External API failures must be handled gracefully without crashing the main application.

4.6.3 Processing Logic

- **Input:** Soil Card ID / Location.
- **Processing:** Authenticate with Gov API -> Fetch XML/JSON data -> Map to internal UI fields.
- **Output:** Fertilizer recommendations and Scheme details.

4.7 Notifications, Alerts, and Digital Records

4.7.1 Description

An asynchronous communication layer that keeps users informed of critical events and maintains an immutable history of all actions for transparency and trust.

4.7.2 Functional Requirements

- **FR-7.1:** The system shall send real-time notifications (Push/SMS/Email) for:
    - Order Confirmation/Cancellation.
    - Booking Approvals.
    - Significant Price drops in the Mandi Dashboard.

- **FR-7.2:** The system shall maintain a **Transaction History** log for every financial exchange, accessible to both parties indefinitely.
- **FR-7.3:** Users shall be able to customize their notification preferences (e.g., opt-out of email marketing).

4.7.3 Processing Logic

- **Input:** Event Triggers (e.g., OrderStatus = 'Shipped').
- **Processing:** Identify affected User ID -> Retrieve Contact Info -> Dispatch Message via Notification Service -> Log in 'Notifications' Table.
- **Output:** User receives alert on device; Audit trail updated.

5. Non-Functional Requirements

5.1. Performance & Scalability

- **Latency:** The Real-Time Mandi Price Dashboard shall display filtered results within 160 seconds on a 3G network connection.
- **Concurrency:** The backend cloud architecture must support up to 5,000 concurrent active users during peak harvest seasons without service degradation.
- **Throughput:** The integrated payment gateway must handle at least 10 transactions per second (TPS) to accommodate bulk B2B settlements.

5.2. Availability & Reliability

- **Operational Uptime**: The platform shall maintain 99.5% availability, excluding scheduled maintenance windows.
- **Graceful Degradation:** If government APIs (eNAM/Soil Health Card) are unreachable, the system shall display the last cached data with a "Time-Since-Update" timestamp rather than an error message.
- **Offline Resilience:** The application must allow for 100% of data entry (crop listings/equipment bookings) to be completed offline, with a background synchronization process triggered as soon as signal is detected.

5.3. Security & Privacy

- **Data Protection:** All user PII (Personally Identifiable Information), including Aadhaar and phone numbers, must be encrypted at rest using AES-256.
- **Authentication:** OTP-based login sessions shall expire after 30 minutes of inactivity to prevent unauthorized access on shared devices.
- **Auditability:** Every financial transaction must be recorded on a decentralized or private ledger to ensure an immutable, tamper-proof audit trail.

5.4. Usability & Accessibility

- **Language Support:** The UI must be available in Hindi and English at launch, with localized dialects added during the pilot phase.
- **Accessibility:** At least 80% of core functions (Selling, Renting, Pricing) must be accessible via voice commands to assist users with low digital literacy.
- **Resource Footprint:** The Android application package (APK) size should not exceed certain limits to accommodate low-storage devices common in rural areas.

6. Software Design

This section describes the internal design structure of the KisanSetu platform. The design is structured to ensure scalability, maintainability, security, and efficient performance under both online and offline conditions. The platform adopts a modular and service-oriented approach to bridge the rural “phygital” divide by integrating digital infrastructure with physical agricultural ecosystems.

6.1. System Architecture

The KisanSetu platform utilizes a **Microservices Architecture** designed to provide flexibility, scalability, and seamless integration with government agricultural systems.

The architecture is logically divided into three primary layers:

6.1.1. Presentation Layer

The Presentation Layer represents the user-facing interface of the platform. It is designed specifically for rural usability and accessibility.

- The interface supports multilingual interaction to improve regional accessibility.
- The layer handles user input validation before forwarding requests to backend services.
- Offline-first design principles allow users to enter data even without active internet connectivity.

This layer communicates with backend services using secure REST-based API calls over encrypted channels.

6.1.2. Application and Integration Layer

The Application & Integration Layer acts as the core operational component of the platform. It processes user requests, applies business logic, and integrates with external systems.

Key components include:

- **API Gateway:** Acts as a centralized request router and security checkpoint. It validates tokens, routes requests to appropriate microservices, and ensures load balancing.
- **User Management Service:** Handles registration, authentication, profile updates, and role management.
- **Crop Listing Service:** Manages produce listings, price discovery, and marketplace interactions.
- **Equipment Sharing Service:** Controls equipment registration, availability tracking, and rental pricing.
- **Booking Management Service:** Handles reservation lifecycle including request, confirmation, cancellation, and completion.
- **Notification Service:** Sends alerts and confirmations to users during booking and transaction processes.

The platform integrates with external government ecosystems such as:

- e-NAM (National Agriculture Market) for mandi price updates
- Soil Health Card databases for soil-related recommendations

This layer ensures interoperability and transparent agricultural data exchange.

6.1.3. Data and Persistence Layer

The Data & Persistence Layer ensures structured storage, secure access, and efficient synchronization of information.

The platform adopts a **dual-database approach**:

1. **Cloud Database**
    - Stores centralized data such as listings, bookings, analytics, and transaction history.
    - Supports AI-driven analytics and price trend evaluation.
    - Enables real-time synchronization across users.

2. **Local SQLite Database**
    - Stores temporary data during offline periods.
    - Ensures 100% functionality for data entry even without network connectivity.
    - Automatically synchronizes with cloud storage once connectivity is restored.

This design ensures resilience, reliability, and uninterrupted service delivery.

6.2. Detailed Design

6.2.1. Class Diagram

6.2.1.1 Abstract class - User

«abstract»
User
-userId : int
-name : String
-email : String
-passwordHash : String ← corrected (bcrypt hash)
-phone : String
-role : Enum {Farmer, Buyer, EquipmentOwner, Admin}
+login() : void
+logout() : void
+updateProfile() : void

6.2.1.2 Sub Classes - Farmer, Buyer, Equipment Owner, Admin

Buyer
-businessType : String
+browseListings() : void
+placeOrder() : void
+confirmReceipt() : void

Farmer
-landArea : float -location : String
+createCropListing() : void +updateCropListing() : void +bookEquipment() : void

EquipmentOwner
-businessName : String
+addEquipment() : void
+updateEquipment() : void
+approveBooking() : void

Admin
-adminLevel : int
+moderateListing() : void
+resolveDispute() : void
+manageUsers() : void

6.2.1.3 Core Entity Classes

CropListing
-cropId : int
-cropName : String
-variety : String
-quantity : int
-basePrice : double
-harvestDate : Date
-listingStatus : String
+publishListing() : void
+updateStatus() : void

Equipment
-equipmentId : int
-equipmentType : String
-model : String
-usageHours : int
-hourlyRate : double
-status : String
-location : String
+checkAvailability() : void
+updateStatus() : void

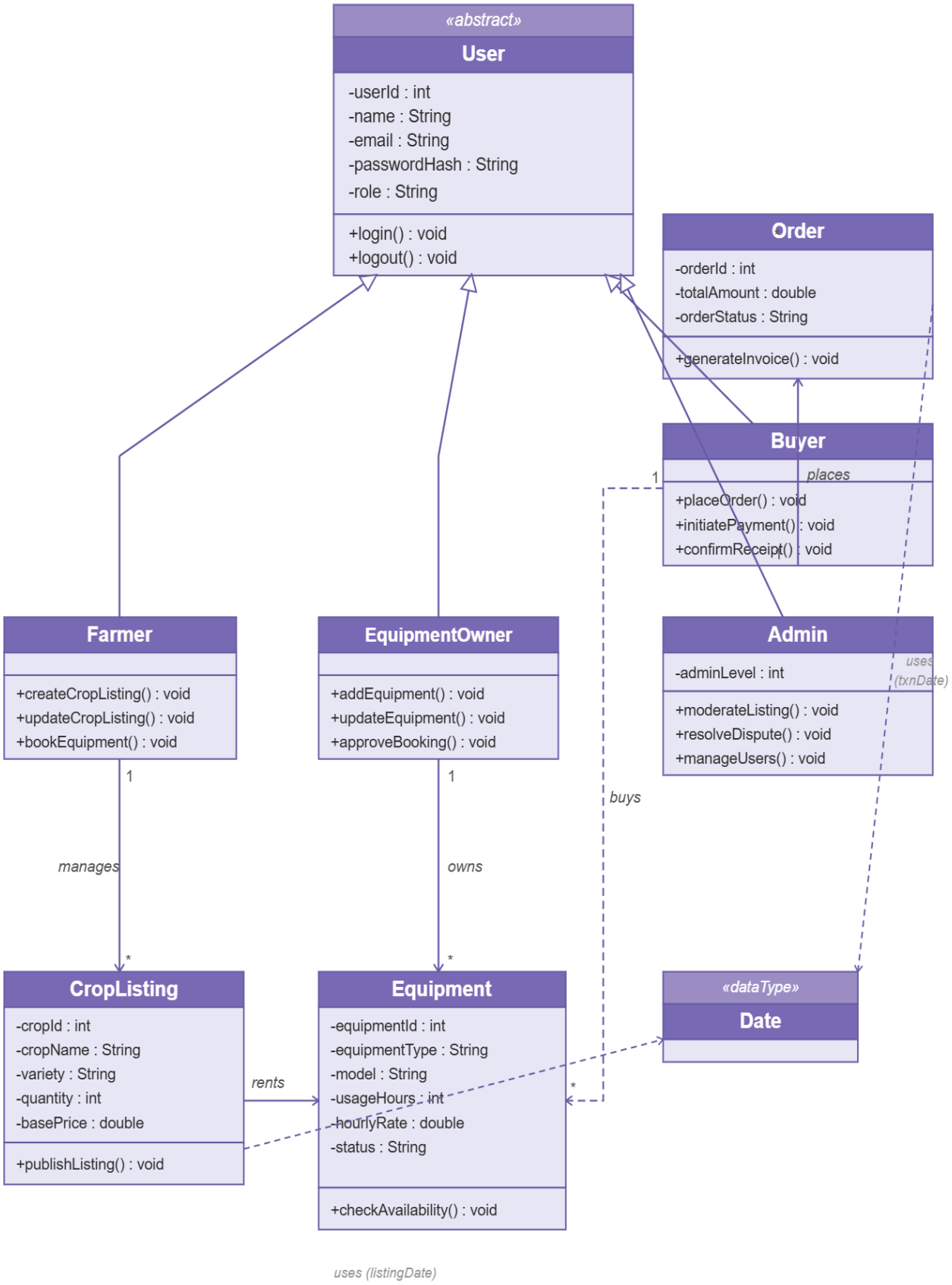
Order
-orderId : int
-orderDate : Date
-totalAmount : double
-orderStatus : String
+generateInvoice() : void
+updateStatus() : void

Booking
-bookingId : int
-startDate : Date
-endDate : Date
-totalCost : double
-bookingStatus : String
+calculateCost() : void
+generateReceipt() : void

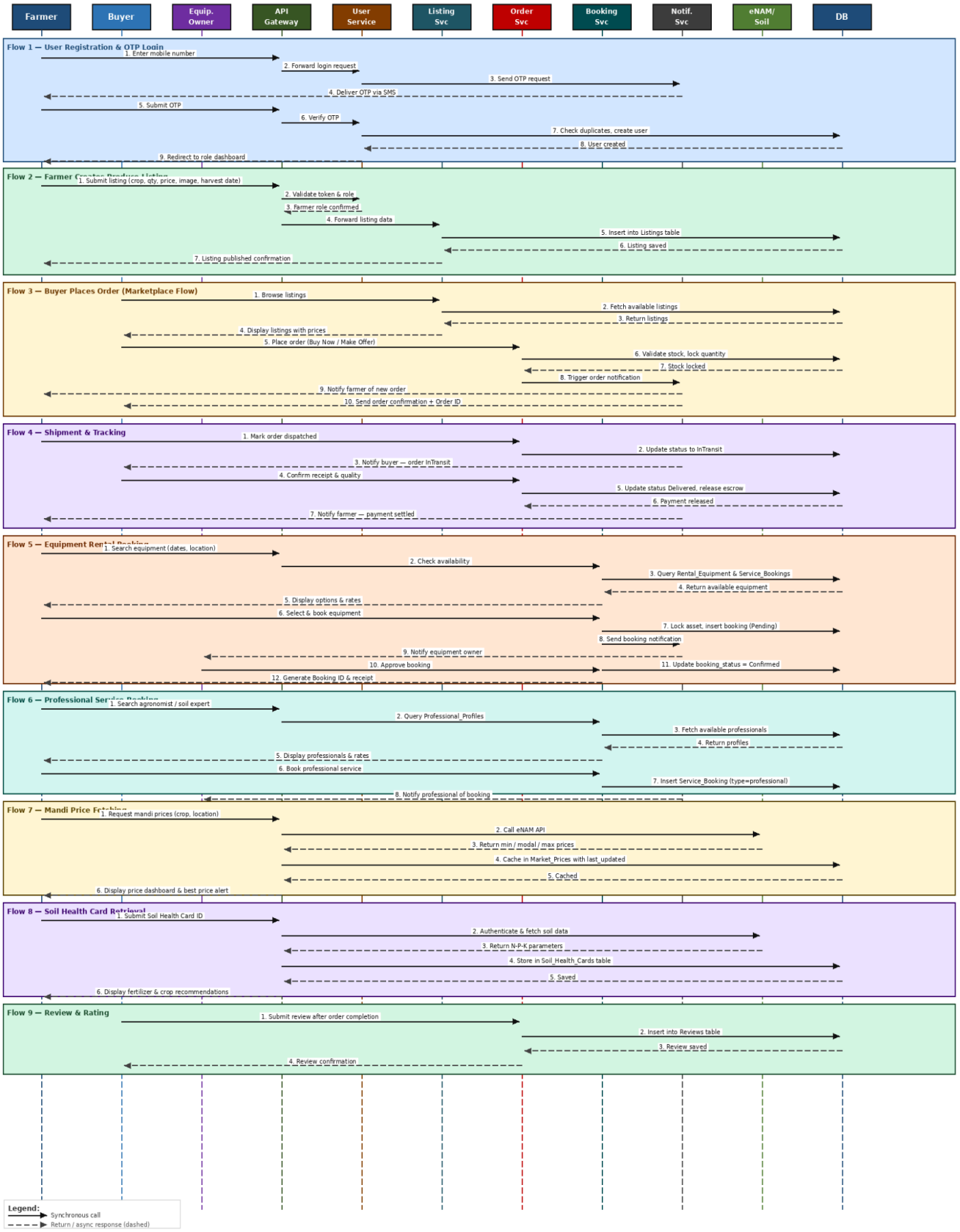
6.1.2.4 Class Relationships

From	Type	To	Multiplicity
Farmer, Buyer, EquipmentOwner, Admin	Inheritance (extends)	User	—
Farmer	Association (manages)	CropListing	1 Farmer → many Listings
Buyer	Association (places)	Order	1 Buyer → many Orders
EquipmentOwner	Association (owns)	Equipment	1 Owner → many Equipment
Buyer / Farmer	Association (rents)	Equipment via Booking	Many → Many via Booking
Order	Dependency (uses)	Date	transactionDate
CropListing	Dependency (uses)	Date	listingDate

6.1.2.5 Class Diagram



6.2.2. Sequence Diagram



6.3. Database Schema

The database schema of the KisanSetu platform is designed to maintain relational integrity, support transactional operations, and ensure efficient data management. The schema follows a normalized relational structure with clearly defined primary keys, foreign keys, and indexing mechanisms.

6.3.1. Database Schema Overview

Table Name	Primary Key	Foreign Key(s)	Main Attributes	Purpose
Users	user_id	—	first_name, last_name, phone_number (unique), email, user_type, date_joined	Stores user account details
Addresses	address_id	user_id → Users	street_address, city, state, pincode, is_primary	Stores user addresses
Phygital_Hubs	hub_id	manager_user_id → Users	hub_name, pincode_served, latitude, longitude	Stores regional service hubs
Products	product_id	—	product_type, product_name, category, unit_of_measure	Stores product master data
Listings	listing_id	farmer_user_id → Usersproduct_id → Products	quantity_available, base_price_per_unit , listing_status	Stores farmer produce listings
Orders	order_id	buyer_user_id → Usersdelivery_address_id → Addresses	order_date, total_amount, order_status	Stores buyer orders
Order_Items	order_item_id	order_id → Orderslisting_id → Listings	quantity_ordered, price_per_unit	Stores items within an order

Rental_Equipment	equipment_id	owner_user_id → Usershub_id → Phygital_Hubs	name, rental_rate_per_hour	Stores rental equipment data
Professional_Profiles	professional_id	professional_id → Users	expertise, hourly_rate, years_experience	Stores professional service data
Service_Bookings	booking_id	client_user_id → Users	service_type, resource_id, time_slot, total_cost	Stores equipment/professional bookings
Market_Prices	price_id	product_id → Products	mandi_name, price_date, min_price, max_price	Stores mandi price records
Soil_Health_Cards	card_id	farmer_user_id → Users	test_date, lab_name, soil_parameters	Stores soil test data
Reviews	review_id	reviewer_user_id → Users	reviewed_entity_id, entity_type, rating	Stores ratings and feedback
Notifications	notification_id	user_id → Users	notification_type, message, is_read, delivery_channel, created_at	Stores all system notifications and user alert preferences for Push/SMS/Email delivery.
HarvestBatch	batch_id	farmer_user_id → Users, listing_id → Listings	crop_type, quantity, storage_location, harvest_date, expiry_date, shelf_life_days, batch_status	Stores individual harvest batch entries for inventory tracking, shelf-life estimation and expiry alerts

6.3.2. Database Design Characteristics

The database design of the KisanSetu platform follows a normalized relational model to ensure data consistency, integrity, and scalability.

Key characteristics include:

- **Normalized Structure:** Each table represents a distinct entity (Users, Listings, Orders, etc.) to reduce redundancy and maintain logical data separation.
- **Primary Keys:** Every table uses a unique primary key for record identification.
- **Foreign Keys:** Relationships between tables are enforced using foreign key constraints to maintain referential integrity.
- **Data Constraints:** ENUM, UNIQUE, NOT NULL, and CHECK constraints ensure data validity and controlled attribute values.
- **Indexing:** Frequently accessed fields such as pincode and category are indexed to improve query performance.
- **Scalability:** The modular structure allows future expansion without major schema modifications.

6.3.3. Relational Integrity Overview

Relational integrity ensures that relationships between entities remain valid and consistent.

The schema maintains the following relationships:

- **One-to-Many:**
  - One User → Multiple Listings
  - One Order → Multiple Order_Items
  - One Shipment → Multiple Tracking Records
  - One Product → Multiple Market Price Records

- **One-to-One:**
  - One Order → One Shipment
  - One Professional Profile → One User

- **Application-Level Relationship:**
  - Service_Bookings links conditionally to Rental_Equipment or Professional_Profiles based on service_type.

Foreign key constraints prevent invalid references and ensure that no orphan records are created. This guarantees consistency across marketplace, logistics, and service operations.

6.4. Security Design

Security is an essential component of the KisanSetu platform architecture. The platform manages sensitive user information, financial transactions, and

agricultural market data; therefore, comprehensive security mechanisms are incorporated to ensure confidentiality, integrity, and availability of information.

The security design is implemented at multiple levels, including user authentication, access control, encrypted communication, and secure data storage. These mechanisms collectively safeguard the platform against unauthorized access, data breaches, and misuse of system resources.

6.4.1. Authentication and Authorization Mechanisms

Authentication mechanisms verify the identity of users accessing the platform, while authorization mechanisms ensure that users are granted access only to permitted functionalities based on their assigned roles.

6.4.1.1. OTP-Based Authentication

The platform implements a mobile number–based authentication mechanism using One-Time Password (OTP) verification.

- During registration or login, the user enters their registered mobile number.
- The system generates and sends a time-bound OTP to the provided number.
- Access is granted only after successful OTP validation.

This approach eliminates dependency on complex password management and enhances usability for rural users. Since access is linked directly to possession of the registered mobile number, unauthorized login attempts are significantly reduced.

6.4.1.2. Role-Based Access Control (RBAC)

The platform supports multiple user roles, including Farmer, Buyer, and Equipment Owner. Role-Based Access Control (RBAC) ensures that system functionalities are restricted according to the assigned role.

- Farmers can create crop listings and initiate equipment bookings.
- Buyers can browse listings and interact with sellers.
- Equipment Owners can register assets and manage booking requests.

Upon successful authentication, the system verifies the user's role and dynamically enables only the relevant interface components. This mechanism prevents privilege misuse and protects sensitive operations from unauthorized modification.

6.4.1.3. Session Management

To prevent unauthorized access, particularly on shared devices, the platform incorporates session management controls.

- A secure session is generated after successful login.
- If no activity is detected for a period of 30 minutes, the session automatically expires.
- Users are required to re-authenticate to continue accessing the platform.

This measure reduces the risk of accidental data exposure and strengthens operational security.

6.4.2. Security Protocols

Security protocols define the technical safeguards implemented to protect stored data and communication channels within the platform.

6.4.2.1. Data Encryption

Sensitive Personally Identifiable Information (PII), including Aadhaar numbers and mobile numbers, is encrypted before being stored in the database.

- The platform utilizes AES-256 encryption standards.
- Encryption at rest ensures that stored data remains unreadable without authorized decryption keys.

This mechanism protects user information even in the event of unauthorized database access.

6.4.2.2. Secure Communication

All communication between the Presentation Layer and the Application & Integration Layer is conducted over secure channels.

- HTTPS protocol is used for all data transmission.
- Transport Layer Security (TLS) encrypts data during transmission.

This prevents interception, unauthorized monitoring, and data manipulation during network communication.

6.4.2.3. Password Hashing

Wherever credential storage is required for administrative functions, passwords are not stored in plain text.

- Secure hashing algorithms such as bcrypt are used.
- Hashing transforms passwords into irreversible encrypted formats.

This ensures that original credentials cannot be retrieved even if database access is compromised.

6.4.2.4. Immutable Audit Trail

To ensure transparency and accountability, the platform maintains a tamper-resistant audit log of all financial and booking transactions.

- Each transaction record includes timestamp, user identification, and transaction details.
- Once recorded, entries cannot be modified or deleted.

The audit trail supports dispute resolution, regulatory compliance, and fraud prevention.

7. Traceability

Traceability ensures that every requirement defined in the earlier sections of this Software Requirements Specification (SRS) is properly addressed in the system design. It establishes a clear relationship between requirements and corresponding design components, ensuring that no requirement is left unimplemented.

Requirement traceability is important for:

- Verifying completeness of system design
- Supporting impact analysis during future modifications
- Facilitating validation and testing
- Ensuring alignment between requirements and implementation

The Requirement-to-Design Mapping table presented below provides structured linkage between Functional Requirements (FR), Non-Functional Requirements (NFR), and their associated design artifacts.

I. Design ID Legend

Design ID Prefix	Design Artifact Category	Reference Section in SRS
D-AR	Architecture & System Layers (Microservices, Frontend, Backend)	Section 6.1 (System Architecture)
D-CL	Class Diagrams	Section 6.2.1 (Class Diagram)
D-SQ	Sequence Diagrams	Section 6.2.2 (Sequence Diagram)
D-DB	Database Schema & Data Dictionary	Section 6.3 (Database Schema) & Appendix B

D-SEC	**Security Design (Authentication, Protocols, Audit Logs)**	**Section 6.4 (Security Design)**
D-AM	**Analysis Models (ERD, State Transition, Decision Tables)**	**Appendix B (Analysis Models)**

7.1. Requirement-to-Design Mapping

7.1.1 Functional Requirements (FR) Traceability Matrix

Requirement ID	Requirement Name	Design Artifacts (IDs & Descriptions)
FR-1.1 to FR-1.7	**Role-Based User Access and Management**	**D-CL-01, D-SQ-01, D-DB-01, D-SEC-01:** Class Diagram (User hierarchy + `bcrypt` password hash); Sequence Diagram (Registration & OTP Login); DB Schema (Users, Addresses); Security Design (RBAC, OTP Auth, 30 min Session, `bcrypt` Hashing).
FR-2.1 to FR-2.5	**Real-Time Mandi Price Dashboard**	**D-SQ-07, D-DB-08, D-AM-03:** Sequence Diagram (Price Fetching Flow); DB Schema (Market_Prices: mandi, date, min/max price); Decision Table (Appendix B: Price Alerts - Urgent Push / Regional In-App / Dashboard Only).
FR-3.1 to FR-3.5	**Direct Produce Listing and Transaction System**	**D-CL-02, D-SQ-02, D-SQ-03, D-DB-04, D-AM-02:** Class Diagram (CropListing, Farmer/Buyer relationships); Sequence Diagrams (Create Listing,

		Place Order, Shipment/Tracking); DB Schema (Listings, Orders, Order_Items); State Transition Diagram/Table (Appendix B: Draft → Listed → ... → Settled).
FR-4.1 to FR-4.5	**Agricultural Equipment Rental and Booking Module**	**D-CL-03, D-SQ-05, D-DB-07, D-AM-01:** Class Diagram (Equipment, EquipmentOwner, Booking); Sequence Diagram (Rental Booking Flow); DB Schema (Rental_Equipment, Service_Bookings); Entity Relational Diagram (Appendix B, Fig 7).
FR-5.1 to FR-5.4	**Inventory and Post-Harvest Management Support**	**D-AR-01, D-SQ-02, D-DB-04:** Application Layer (Crop Listing Service); Sequence Diagram (Listing creation, includes batch/quantity tracking); DB Schema (Listings table fields support inventory, shelf-life, and expiry alerts).
FR-6.1 to FR-6.4	**Government Service Integration Interface**	**D-AR-01, D-SQ-08, D-DB-09:** Application Layer (Integration with e-NAM/Soil Health Card APIs); Sequence Diagram (Soil Health Card Retrieval); DB Schema (Soil_Health_Cards); Data Flow Diagram (Fig 4: External API integration with failure handling).
FR-7.1 to FR-7.3	**Notifications, Alerts and Digital Records**	**D-AR-01, D-SEC-03:** Application Layer (Notification Service for Order/Booking/Price alerts via Push/SMS/Email); Security Design (Immutable

		Audit Trail for financial exchanges); Full System Sequence (covers all notification triggers).

7.1.2 Non-Functional Requirements (NFR) Traceability Matrix

Requirement ID	Requirement Name	Design Artifacts (IDs & Descriptions)
NFR-5.1	**Performance and Scalability**	**D-AR-01:** Microservices Architecture (with API Gateway/load balancing, Section 6.1); Cloud DB supporting 5,000 concurrent users; Payment Gateway targeting 10 TPS for B2B settlements.
NFR-5.2	**Availability and Reliability**	**D-AR-01, D-AM-04:** Dual-Database (Cloud DB + Local SQLite for 100% offline resilience, Section 6.1.3); System Resilience Diagram (Fig 5); Connectivity Monitor Logic (Appendix B, Fig 8: 2G/3G MQTT fallback); 99.5% uptime with graceful API degradation.
NFR-5.3	**Security and Privacy**	**D-SEC-01, D-SEC-02, D-SEC-03:** Security Design (Section 6.4): OTP Auth, RBAC, 30-min Session Timeout, AES-256 Encryption for PII, HTTPS/TLS, `bcrypt` Hashing, Immutable Audit Trail.
NFR-5.4	**Usability and Accessibility**	**D-AR-01:** Presentation Layer (Section 6.1.1): Multilingual (Hindi/English); Voice-first interface (80%

		core functions via voice); Offline-first design; Icon-driven navigation (for low digital literacy); Lightweight APK (for mid-to-low-range Android).

Appendix A: Glossary

1. Agricultural & Domain Terms

- **Agri-Ecosystem:** The complex network of farmers, buyers, service providers, and government bodies interacting within the agricultural sector.
- **B2B (Business-to-Business):** Transactions where farmers sell in bulk to entities like wholesalers, food processors, or retail chains.
- **B2C (Business-to-Consumer):** Transactions where farmers sell directly to individual end-users or households.
- **e-NAM (Electronic National Agriculture Market):** A pan-India electronic trading portal that networks existing APMC mandis to create a unified national market for agricultural commodities.
- **FPO (Farmer Producer Organization):** A legal entity formed by primary producers (farmers) to provide better bargaining power, access to credit, and shared resources.
- **Mandi:** A traditional Indian marketplace or "yard" where agricultural produce is traded, usually regulated by a market committee (APMC).
- **Post-Harvest Loss:** The measurable quantitative and qualitative loss of a crop from the time of harvest until it reaches the consumer.
- **Smallholder Farmer:** A farmer cultivating a small landholding (typically 1–2 hectares or 1–5 acres), often characterized by limited access to technology and credit.
- **Soil Health Card (SHC):** A government scheme providing farmers with crop-wise recommendations of nutrients and fertilizers required for their individual farms.

2. Technical & Platform Terms

1. **API (Application Programming Interface):** A set of rules that allows KisanSetu to "talk" to and fetch data from other systems like e-NAM or weather services.
2. **Asset-Sharing Model:** A "rental" economy where farmers can book expensive machinery (tractors, drones) they don't own on a pay-per-use basis.
3. **Micro-Entrepreneurs / Field Agents:** Local tech-savvy individuals who help onboard farmers and manage physical collection points.
4. **P2P (Peer-to-Peer):** A decentralized model where two individuals (e.g., two farmers or a farmer and a neighbor) interact directly without a third-party intermediary.
5. **Phygital:** A combination of "Physical" and "Digital." It refers to providing digital services through physical locations (hubs) and human assistance (agents).

6. **SRS (Software Requirements Specification):** A document that describes what the software will do and how it will be expected to perform.
7. **UI/UX (User Interface / User Experience):** The visual design (UI) and the overall feeling or ease of use (UX) of the KisanSetu.
8. **Voice-First Interface:** A design approach that allows users to interact with the platform primarily through speech, crucial for users with low digital literacy.

Appendix B - Analysis Models

● Entity Relational Diagram

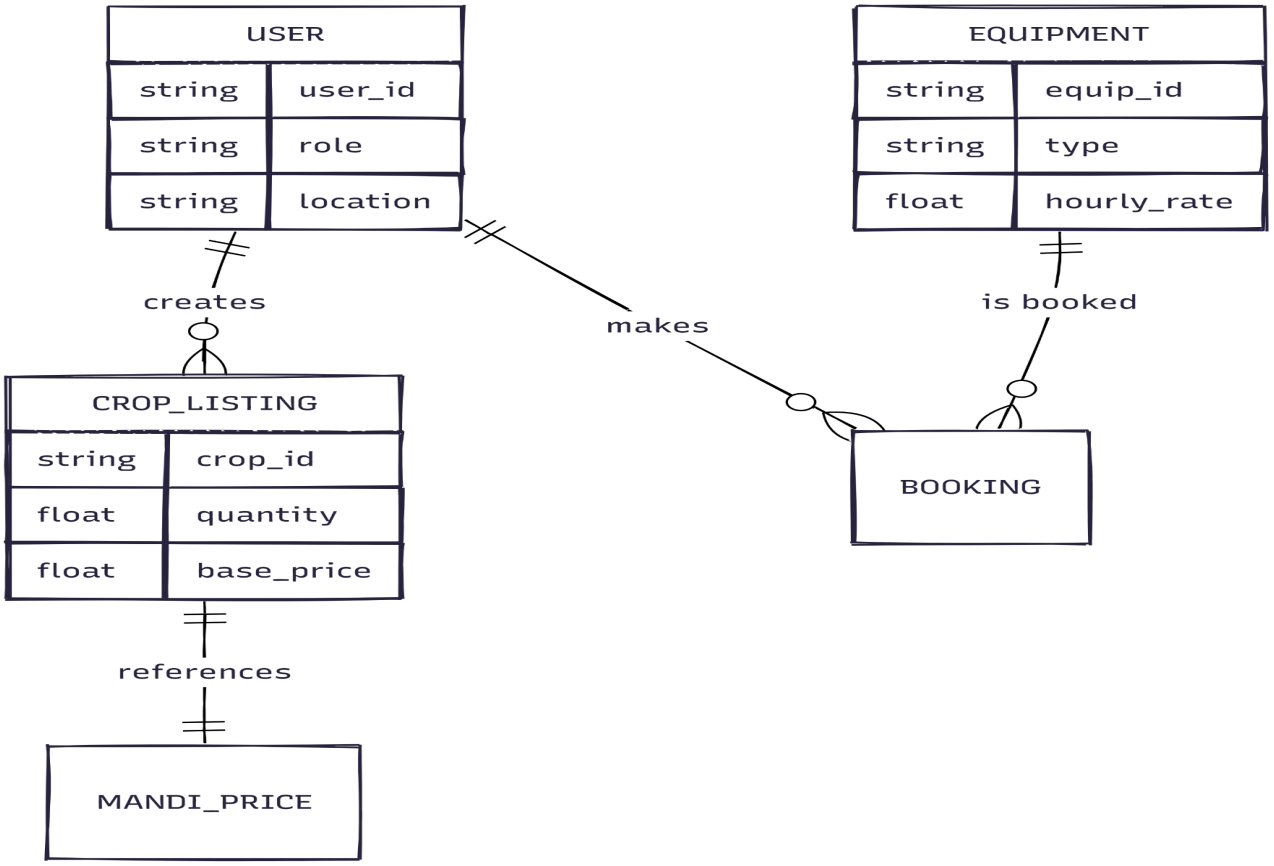


Fig 7 : Database structure required to support the platform.

● State Transition Table

This table defines the logic for our **Crop Listing and Negotiation Module**. It explains how the system transitions between states based on user actions.

Current State	Event (Trigger)	Next State	Action Taken
Draft	Farmer clicks "Publish"	**Listed**	Validate photo/input and set base price.

Listed	Buyer places Bid	**Negotiation**	Notify Farmer of new offer/bid.
Negotiation	Farmer accepts Price	**Locked**	Reserve stock and disable other active bids.
Locked	Quality Check Fails	**Listed**	Re-open listing; Notify Buyer of cancellation.
Delivered	Buyer confirms Quality	**Processing**	Initiate escrow fund release via UPI/Bank.

- ## Decision Table: Mandi Price Alerts

This table provides the logic for the **Real-Time Mandi Price Dashboard** alerts.

Condition	**Rule 1 (Urgent)**	**Rule 2 (Regional)**	**Rule 3 (Informational)**
Price > 10% average?	Yes	Yes	No
Distance < 50km?	Yes	No	Yes
Alert Action	**Push Notification**	**In-App Notification**	**Dashboard Only**

- ## Communication Logic

To visualize the **Connectivity Monitor** logic mentioned in our technical resilience section.

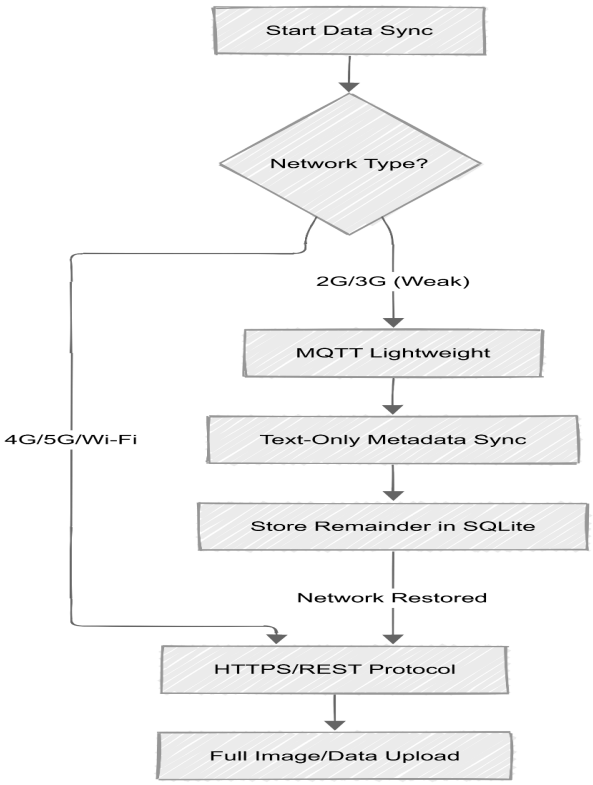


Fig 8 : Connectivity Monitor logic

- ## Data Dictionary

This defines the technical storage requirements for our **Database Structure**.

Field Name	Data Type	Constraints	Description
user_id	String	Primary Key	Unique system identifier for all users.
base_price	Float	Not Null	Minimum price a farmer will accept for a crop.
hourly_rate	Float	Not Null	Rental cost for machinery per hour.
location	String	Mandatory	Geographic data for price discovery/logistics.

Appendix C - To Be Determined List

This table documents the key technical and business decisions that are 'To Be Determined' (TBD) within the scope of the project's Software Requirements Specification (SRS), along with their current status, source reference in the SRS document, and the specific rationale for why a final decision cannot yet be made.

ID	Item	Source Reference	Reason for TBD Status - Detailed Elaboration
TBD-01	**Payment Gateway Vendor**	Section 3.3 (…	The final selection of the payment gateway partner is contingent upon a comprehensive cost-benefit analysis. The core requirement is to identify the provider that offers the most favorable transaction fee structure, particularly for low-value, high-volume transactions typical of UPI-based settlements in rural agricultural markets. This selection directly impacts the project's operational budget and the eventual fee passed on to the end-user.
TBD-02	**Voice-First NLP Engine**	Section 2.3.1 …	A crucial technical evaluation is required to determine the optimal Natural Language Processing (NLP) engine for the voice-first interface. The decision is binary: Bhashini or Google Speech-to-Text. The determining factor is the relative accuracy and robustness of each platform in recognizing and accurately transcribing the specific, nuanced local dialects and regional variations spoken by farmers in the designated pilot region. Successful adoption hinges on the quality of this

			engine.
TBD-03	**Soil Health Parameters Data Source**	Section 3.3 (…	The project requires real-time or near real-time access to twelve specific soil health parameters. The TBD status relates to the integration method with government agricultural laboratories. Confirmation is urgently required from these labs to ascertain if they can expose all 12 parameters via a reliable, standardized real-time API feed, or if the system will be required to build a facility for manual, periodic bulk upload of data, which would introduce latency and potential data-entry errors.
TBD-04	**"Last Mile" Logistics Partner**	Section 1.3.1 …	The strategy for "B2C local delivery" in the agricultural supply chain remains unresolved. The decision point is whether the project will leverage existing internal field agents/personnel to perform the last-mile deliveries or if it will establish a formal partnership with an established third-party logistics (3PL) provider specializing in rural delivery, such as Delhivery or Ecom Express. This choice affects operational scale, delivery speed, and cost per delivery.

TBD-05	**Blockchain Framework**	Section 3.3 (…	The implementation of a distributed ledger for ensuring "tamper-proof records" is mandated. The TBD involves selecting the appropriate underlying framework. The trade-off is between the full feature set and enterprise support of a solution like Hyperledger (offering robust governance) versus the lower overhead and simplified deployment of a proprietary, simplified private ledger. The final choice must balance the security/audit requirement with the necessity of low-cost, sustainable operation.
TBD-06	**Mandi Price Data Refresh Frequency**	Section 4.2.2 …	The system's utility for farmers depends on the timeliness of agricultural commodity price information from Mandis (local markets). The TBD status is dependent on the capabilities of the eNAM (electronic National Agriculture Market) APIs. The engineering team must confirm whether the eNAM interface allows for a frequent refresh rate (e.g., every 4 hours) to provide intra-day price movements, or if the system will be constrained to relying only on the 24-hour daily price averages, which would limit the system's real-time decision-support value.